

WORK SHEET - 11

ELLIPSE & HYPERBOLA

Q.1 $\frac{\sqrt{3}}{2}$

Sol. $\frac{dy}{dx} = 1 - 3x^2|_{(-1, 0)} = -2$

$\therefore a = -2$

Also $(-1, 0)$ lies on line

$0 = b - a \quad \therefore b = -2$

E: $\frac{x^2}{16} + \frac{y^2}{4} = 1$

$e^2 = 1 - \frac{4}{16} = 1 - \frac{1}{4} = \frac{3}{4}$

$e = \frac{\sqrt{3}}{2}$. Ans.

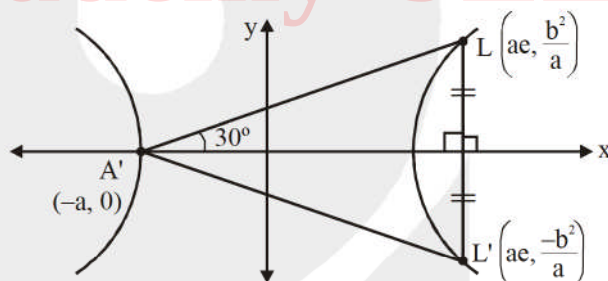
Q.2 $\frac{\sqrt{3}+1}{\sqrt{3}}$

Sol. Now, $\tan 30^\circ = \frac{b^2}{a(ae+a)}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{b^2}{a^2(1+e)}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{(e^2-1)}{(1+e)} \Rightarrow e-1 = \frac{1}{\sqrt{3}}$

$\Rightarrow e = \frac{\sqrt{3}+1}{\sqrt{3}}$ Ans.

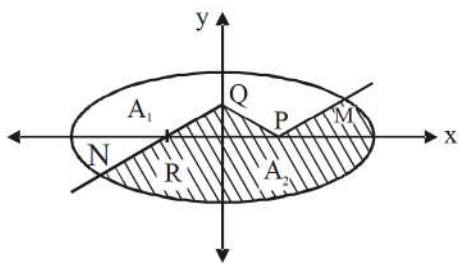


Q.3 $A_1 : A_2 = 7 : 13$

Sol. 40517/elli/MORE Area of an ellipse = $\pi ab = \frac{20}{3}$, Area of triangle PQR = 1

$\therefore A_1 = \frac{7}{3}; A_2 = \frac{13}{3}$

[By symmetry area M = area N]



Q.4 $x^2 - \frac{y^2}{8} = 1$

Sol. Let the equation of variable circle be

$$(x - h)^2 + (y - k)^2 = r^2 \quad \dots\dots(1)$$

Now, according to question

$$CC_1 = r + 4$$

$$CC_2 = r + 2$$

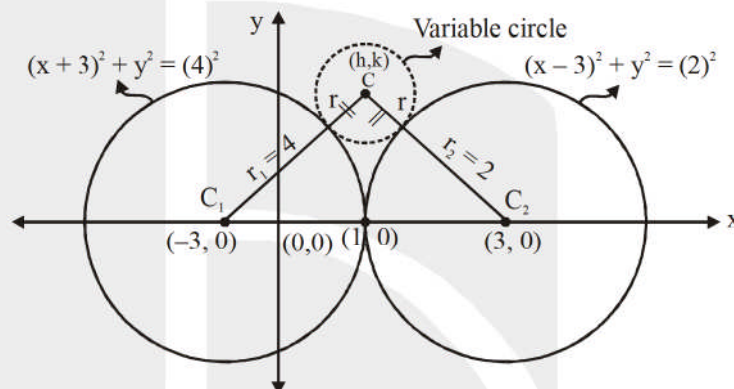
$$\therefore |CC_1 - CC_2| = 2$$

$\Rightarrow$ The locus of $P(h, k)$ is a hyperbola with foci $C_1(-3, 0)$ & $C_2(3, 0)$ and transverse axis of length 2.

$$\text{Also, } 2a \cdot e_h = 6 \Rightarrow e_h = \frac{6}{2} = 3$$

$$\text{Also, } e_h^2 = 1 + \frac{b^2}{a^2} \Rightarrow 9 - 1 = \frac{b^2}{1} \Rightarrow b^2 = 8$$

So, equation of hyperbola is $\frac{x^2}{1} - \frac{y^2}{8} = 1$ **Ans.**



Q.5 $\frac{1}{\sqrt{3}}$

Sol. $P(a \cos \theta, b \sin \theta)$

$$\text{Equation of normal } ax \sec \theta - by \operatorname{cosec} \theta = a^2 e^2$$

$$\therefore Q\left(\frac{ae^2}{\sec \theta}, 0\right) \text{ \& \& } R\left(0, \frac{a^2 e^2}{b \operatorname{cosec} \theta}\right)$$

Now $PQ : QR = 2 : 1$

Using section formula

$$ae^2 \cos \theta = \frac{2(0) + 1(a \cos \theta)}{3}$$

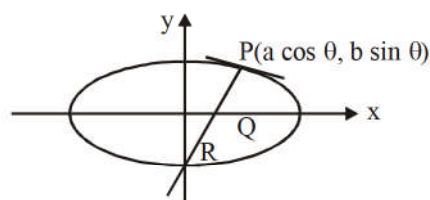
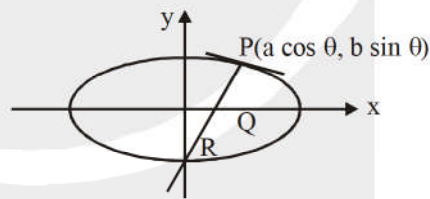
$$\text{Hence } e^2 = \frac{1}{3} \Rightarrow e = \frac{1}{\sqrt{3}} \text{ **Ans.**}$$

Aliter: $P(a \cos \theta, b \sin \theta)$

$$\text{Equation of normal } ax \sec \theta - by \operatorname{cosec} \theta = a^2 e^2$$

$$\therefore Q\left(\frac{ae^2}{\sec \theta}, 0\right) \text{ \& \& } R\left(0, \frac{a^2 e^2}{b \operatorname{cosec} \theta}\right)$$

Put value of PQ and PR in equation



$$9PQ^2 = 4PR^2$$

$$9[(a \cos \theta - ae^2 \cos \theta)^2 + (b \sin \theta - 0)^2] = 4 \left[(a \cos \theta - \theta)^2 + \left(b \sin \theta + \frac{a^2 e^2}{b} \sin \theta \right)^2 \right]$$

$$9a^2 \cos^2 \theta (1 - e^2)^2 + 9b^2 \sin^2 \theta = 4a^2 \cos^2 \theta + 4 \cdot \frac{a^4}{b^2} \sin^2 \theta \quad [\because a^2 e^2 = a^2 - b^2]$$

$$9 \cos^2 \theta \cdot \frac{b^4}{a^2} - 4a^2 \cos^2 \theta = 4 \frac{a^4}{b^2} \sin^2 \theta - 9b^2 \sin^2 \theta$$

$$\therefore \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) [9b^4 - 4a^4] = 0$$

$$\therefore \frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \neq 0$$

$$\therefore 9b^4 = 4a^4 \Rightarrow \frac{b^2}{a^2} = \frac{2}{3}$$

$$1 - e^2 = \frac{2}{3} \Rightarrow e = \frac{1}{\sqrt{3}}$$

Q.6 3

Sol. We have

$$\frac{x^2}{9} - \frac{y^2}{7} = 1 \text{ (equation of hyperbola)}$$

$$\text{Here, } e_h^2 = 1 + \frac{7}{9} = \frac{16}{9} \Rightarrow e_h = \frac{4}{3}$$

$$\text{Also, } \frac{x^2}{25} + \frac{y^2}{b^2} = 1 \text{ (equation of ellipse)}$$

$$\text{Here, } e'^2 = 1 - \frac{b^2}{25} \Rightarrow e' = \sqrt{1 - \frac{b^2}{25}} \text{ (e' = eccentricity of ellipse)}$$

$$\therefore \text{ Foci of ellipse is } (\pm \sqrt{25 - b^2}, 0)$$

As ellipse and hyperbola intersect orthogonally, so they must be confocal (i.e., same foci)

$$\text{So, } \sqrt{25 - b^2} = 4$$

$$\Rightarrow 25 - b^2 = 16 \Rightarrow b^2 = 9$$

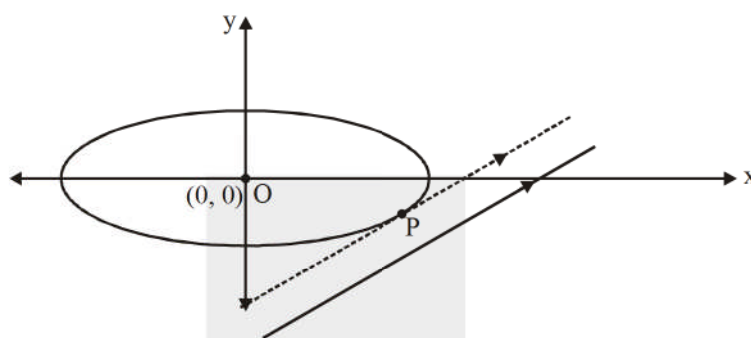
$$\therefore b = 3 \text{ Ans.}$$

Q.7 (2, -1)

Sol. $P(\sqrt{6} \cos \theta, \sqrt{3} \sin \theta)$ lies on ellipse.

Also, tangent at P is parallel to $x - y = 7$

$$\therefore 1 = \frac{-\sqrt{6} \cos \theta}{2\sqrt{3} \sin \theta} \Rightarrow \tan \theta = \frac{-1}{\sqrt{2}}$$



Also, P must lie in the fourth quadrant on ellipse,

$$\text{So, } \sin \theta = \frac{-1}{\sqrt{3}}, \cos \theta = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\text{So, } P(x=2, y=-1)$$

Q.8 $\frac{\pi}{4}$

Sol. Let $P(\theta) = (a \cos \theta, b \sin \theta)$

So, tangent at P to the ellipse is given by

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad \dots (1)$$

$\therefore$ On comparing equation (1) with $\frac{x}{a\sqrt{2}} + \frac{y}{b\sqrt{2}} = 1$,

we get,

$$\cos \theta = \sin \theta = \frac{1}{\sqrt{2}} \text{ (each)}$$

$$\Rightarrow \theta = \frac{\pi}{4} \text{ (Aq.)}$$

Q.9 $x^2 + y^2 = 1$

Q.10 $\left(0, \pm \frac{\sqrt{3}}{2}\right)$

Q.11 $x^2 + y^2 = \frac{5}{4}$

Sol. Let the fixed ellipse whose axes are coordinate axes be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The line

$$y = m'x \pm \sqrt{a^2 m'^2 + b^2} \quad \dots (1)$$

touches the ellipse for infinitely many real values of m .

Hence (1) is identical with

$$y = \frac{-2m}{\sqrt{1-m^2}}x + \frac{1}{\sqrt{1-m^2}} \quad \dots(2)$$

$$\text{So, } m' = \frac{-2m}{\sqrt{1-m^2}}$$

$\therefore$ Using condition of tangency, we get

$$\frac{1}{(1-m^2)} = a^2 \left(\frac{4m^2}{1-m^2} \right) + b^2$$

$$\Rightarrow (4a^2 - b^2)m^2 + (b^2 - 1) = 0,$$

which is true for infinitely many real values of m .

So,

$$b^2 = 1 \text{ and } 4a^2 = b^2$$

$$\text{So, } b^2 = 1 \text{ and } a^2 = \frac{1}{4}$$

$\therefore$ The equation of ellipse

$$\frac{x^2}{1/4} + \frac{y^2}{1} = 1 \quad (\text{vertical ellipse})$$

Now,

$$\frac{1}{4} = 1 - e^2 \Rightarrow e^2 = \frac{3}{4} \Rightarrow e = \frac{\sqrt{3}}{2}$$

$$\text{So, foci} = \left(0, \pm \frac{\sqrt{3}}{2} \right)$$

and equation of auxiliary circle and director circle are $x^2 + y^2 = 1$ and $x^2 + y^2 = \frac{5}{4}$ respectively.

Aliter: In the equation,

$$2mx + y\sqrt{1-m^2} = 1$$

Put $m = \sin\theta$

Equation becomes

$$2 \tan \theta x + y = \sec \theta$$

is a tangent to ellipse.

$$\text{So if } \theta \rightarrow \frac{\pi}{2} \quad m_T \rightarrow \infty$$

$$\Rightarrow x = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1}{2 \sin \theta} = \frac{1}{2} = a$$

$$\text{and } \theta \rightarrow 0 \quad m_T \rightarrow 0$$

$$\Rightarrow y = \lim_{\theta \rightarrow 0} \sec \theta = 1 = b$$

$$a = \frac{1}{2} \text{ and } b = 1$$

∴ Equation of auxillary circle

$$x^2 + y^2 = b^2 \Rightarrow x^2 + y^2 = 1$$

Focii = (0, ±be)

$$e = \sqrt{1 - \frac{1}{2^2}} = \frac{\sqrt{3}}{2}$$

$$\text{Focii} = \left(0, \pm \frac{\sqrt{3}}{2}\right)$$

Director circle

$$x^2 + y^2 = a^2 + b^2 \Rightarrow x^2 + y^2 = \frac{1}{4} + 1 \Rightarrow x^2 + y^2 = \frac{5}{4}$$

Q.12 (B) length of latus rectum equals 20.

(D) distance between foci equals $5\sqrt{5}$

Sol. The hyperbola is $\frac{x^2}{4} - \frac{y^2}{1} = 1$

∴ Here a = 2, b = 1

So, equation of normal at P is

$$2x \cos \theta + y \cos \theta = 5$$

so, A $\left(\frac{5}{2} \sec \theta, 0\right)$ and B(0, 5tanθ)

⇒ Equation of lines drawn from A and B perpendicular to x-axis and y-axis are

$$x = \frac{5}{2} \sec \theta \quad \dots (1)$$

$$y = 5 \tan \theta \quad \dots (2)$$

Let point of intersection of (1) and (2) be (h, k). So,

using $\sec^2 \theta - \tan^2 \theta = 1$

$$\Rightarrow \frac{4x^2}{25} - \frac{y^2}{25} = 1$$

or $\frac{x^2}{25/4} - \frac{y^2}{25} = 1$, which is equation of hyperbola.

$$\text{Now, } e_h^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{25}{25/4} = 5$$

$$\Rightarrow e_h = \sqrt{5}$$

$$\text{and } l(\text{L.R.}) = \frac{2b^2}{a} = \frac{2 \times 25}{5/2} = 4 \times 5 = 20$$

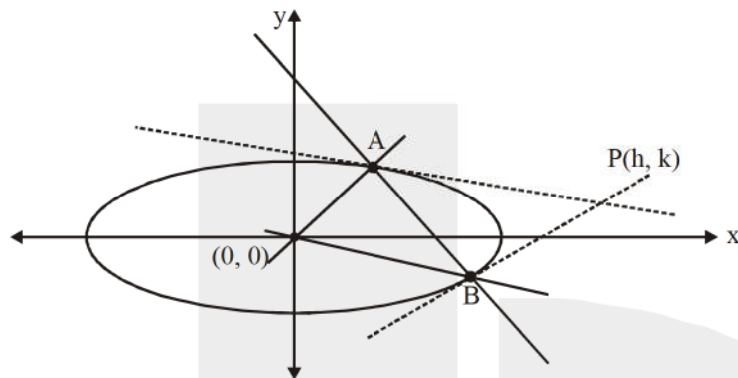
Also, equation of auxiliary circle of hyperbola is $x^2 + y^2 = \frac{25}{4}$.

$$\text{Note that distance between foci} = 2ae_h = 2\left(\frac{5}{2}\right)\sqrt{5} = 5\sqrt{5}$$

Q.13 (A) $x - 2y = 0$ (C) $x + 2y = 0$

Sol. Let the point of intersection of tangents at A and B be P(h, k) then equation of AB is

$$\frac{hx}{4} + \frac{ky}{1} = 1 \quad \dots (1)$$



Now, homogenising the equation of ellipse using (1), we get

$$\frac{x^2}{4} + \frac{y^2}{1} = \left(\frac{hx}{4} + \frac{ky}{1} \right)^2$$

$$\Rightarrow \left(\frac{h^2 - 4}{16} \right) x^2 + (k^2 - 1)y^2 + \frac{2hk}{4} xy = 0 \quad \dots (2)$$

Also, the given combined equation of OA and OB is

$$x^2 + 4y^2 + \alpha xy = 0 \quad \dots (3)$$

$\therefore$ Equation (2) and equation (3) are identical, so

$$\frac{\left(\frac{h^2 - 4}{16} \right)}{1} = \frac{k^2 - 1}{4} = \frac{hk}{2\alpha}$$

$$\Rightarrow (h^2 - 4) = 4(k^2 - 1)$$

$$\Rightarrow h^2 - 4k^2 = 0$$

$$\therefore (h - 2k)(h + 2k) = 0$$

so, locus of P(h, k) is

$$x - 2y = 0 \text{ or } x + 2y = 0 \quad \text{Ans.}$$

Q.14 (A) minimum area of ΔOAB is 6.

(C) minimum length of AB is 5.

Sol. The equation of tangent at P on ellipse is

$$\frac{x}{3} \cos \theta + \frac{y}{2} \sin \theta = 1$$

$$\text{So, } A \left(\frac{3}{\cos \theta}, 0 \right), B \left(0, \frac{2}{\sin \theta} \right)$$

$$\text{Now, an } (\Delta OAB) = \frac{1}{2} \left| \left(\frac{3}{\cos \theta} \right) \cdot \left(\frac{2}{\sin \theta} \right) \right| = \left| \frac{6}{\sin 2\theta} \right| \geq 6.$$

$$\Rightarrow \text{Minimum area of } \Delta OAB = 6 \quad \text{Ans.}$$

$$\begin{aligned}\text{Also, } AB &= \sqrt{9\sec^2\theta + 4\operatorname{cosec}^2\theta} = \sqrt{9(1+\tan^2\theta) + 4(1+\cot^2\theta)} \\ &= \sqrt{13 + (3\tan\theta - 2\cot\theta)^2 + 12} = \sqrt{25 + (3\tan\theta - 2\cot\theta)^2} \geq 5\end{aligned}$$

$\Rightarrow$ Minimum length AB is 5. Ans.

Q.15 (A) $16x - 3y = 0$ (B) $9y^2 = 32x$ (C) $24x + 24y = 19$

Sol. Let $P(h, k)$ be any point on line $3x + 4y = 24$

$\therefore$ The equation of chord of contact is

$$hx + \frac{ky}{4} = 1, \dots (1)$$

where $3h + 4k = 24 \dots (2)$

$\therefore$ From (1) and (2), we get

$$hx + \frac{(24-3h)}{4} \cdot \frac{y}{4} = 1 \quad \left[\because k = \frac{24-3h}{4} \right]$$

$$\Rightarrow h \left(x - \frac{3y}{16} \right) + \left(\frac{3y}{2} - 1 \right) = 0$$

So, point of concurrency is

$$\left(x = \frac{1}{8}, y = \frac{2}{3} \right)$$

Now, verify alternatives.

Q.16 (A) $\alpha + \beta = -4$

(D) Area of $\Delta PAB = 32\sqrt{2}$

Sol. $\frac{x^2}{4} + \frac{y^2}{6} = 1$ and $y^2 = 8x$

Tangent to the ellipse

$$y = mx \pm \sqrt{4m^2 + 6}$$

Tangent to the parabola

$$y = mx + \frac{2}{m}$$

$$\therefore \frac{2}{m} = \pm \sqrt{4m^2 + 6}$$

$$4 = m^2(4m^2 + 6)$$

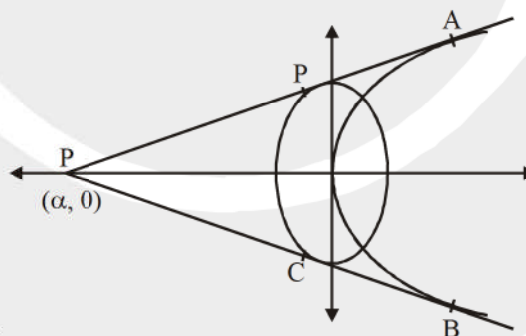
$$2m^4 + 3m^2 - 2 = 0$$

$$(m^2 + 2)(2m^2 - 1) = 0 \Rightarrow m = \pm \frac{1}{\sqrt{2}}$$

$$\therefore y = \frac{x}{\sqrt{2}} + 2\sqrt{2}$$

$$\sqrt{2}y = x + 4 \Rightarrow \alpha = -4, \beta = 0$$

$$\text{Ar}(\Delta PAB) = \frac{S_1^{3/2}}{2|a|} = \frac{(0+32)^{3/2}}{2 \cdot 2} = \frac{32 \cdot 4\sqrt{2}}{4} = 32\sqrt{2}$$



PART-D

Q.17 5

Sol. Clearly locus of centres all such circles will be the tangent to ellipse at point $P(2, \sqrt{3})$.

$$\therefore T: \frac{2x}{8} + \frac{\sqrt{3}y}{6} = 1$$

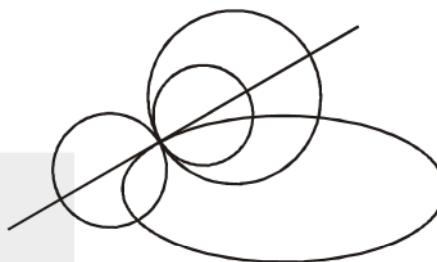
$$\frac{x}{4} + \frac{\sqrt{3}y}{6} = 1$$

$$3x + 2\sqrt{3}y = 12$$

$$\sqrt{3}x + 2y = 4\sqrt{3}$$

$$a = 3, b = 4, c = 3$$

$$\therefore \left| \frac{a+b+c}{2} \right| = 5 \text{ Ans.}$$



Q.18 7

Sol.
$$\frac{\left(\frac{3x-4y+5}{5}\right)^2}{4} + \frac{\left(\frac{4x+3y-10}{5}\right)^2}{2} = 1$$

$$\therefore m = 4, m = 2\sqrt{2}$$

$$\text{Major axis : } 4x + 3y - 10 = 0$$

$$\text{minor axis : } 3x - 4y + 5 = 0$$

$$m \leq p_1 + p_2 \leq M$$

$$2\sqrt{2} \leq \left| \frac{4x+3y-10}{5} \right| + \left| \frac{3x-4y+5}{5} \right| \leq 4$$

$$2\sqrt{2} \leq |x| + |y| \leq 4$$

$$\text{Required area} = 32 - 16 = 16. \text{ Ans.}$$

